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[Programming Project #1](#)

Images of the Russian Empire

Colorizing the Prokudin-Gorskii glass plate collection

Between 1907 and 1915, Sergei Prokudin-Gorskii photographed the Russian Empire in color, decades before color film existed. He did so by exposing three black-and-white plates of the same scene through red, green, and blue filters. Reconstructing the color photo means finding the small pixel shift that aligns the three plates back together. This project builds that alignment pipeline: an exhaustive search over candidate shifts for small images, and a coarse-to-fine image pyramid for the full-resolution scans, where an exhaustive search alone is too slow.



cathedral.jpg · L2, single-scale

dx +2, dy +5 (G) · dx +3, dy +12 (R)

01 · Method

Single-scale exhaustive search

Each glass plate is one tall image, stacked B / G / R from top to bottom. After splitting it into three equal thirds, the green and red channels are each shifted by every candidate (dx, dy) in a ± 15 pixel window and scored against the blue channel. Two scoring metrics were implemented:

L2 / Euclidean distance

```
sqrt(sum((img1 - img2)2))
```

A difference measure — the shift that *minimizes* this is kept.

Normalized cross-correlation

```
dot(img1̄, img2̄), mean-subtracted & unit-normalized
```

A similarity measure — the shift that *maximizes* this is kept.

Each plate also carries its own black border on all four sides. Scoring on the full channel let that border dominate the metric and pulled the search toward the wrong shift, so both channels are cropped by 15% on every side before scoring. The found shift is then applied to the *full, uncropped* channel so no real image content is lost in the final output.

02 • Results

Single-scale results, low-resolution JPEGs

On the three provided low-res images, L2 and NCC agree exactly. I interpreted this as a good sign, since two independently-implemented metrics converging on the same answer is strong evidence the shift is correct.



cathedral

L2 dx +2, dy +5 (G) · dx
 +3, dy +12 (R)
NCC dx +2, dy +5 (G) · dx
 +3, dy +12 (R)



monastery

L2 dx +2, dy -3 (G) · dx
 +2, dy +3 (R)
NCC dx +2, dy -3 (G) · dx
 +2, dy +3 (R)



tobolsk

L2 dx +3, dy +3 (G) · dx
 +3, dy +6 (R)
NCC dx +3, dy +3 (G) · dx
 +3, dy +6 (R)

03 • Method

Coarse-to-fine image pyramid

The full-resolution .tif scans are large enough that a ± 15 window is both too slow to search exhaustively and too narrow to contain the true displacement, which scales up with resolution. The pyramid resolves both problems by recursing on progressively half-sized versions of each channel:

- 1 Downsample both channels by $2\times$ until the image is small — small enough that an exhaustive ± 15 search is both fast and guaranteed to contain the (now much smaller) true shift.
- 2 Solve at that coarsest scale, then double the found shift on the way back up — a shift of N pixels at half-resolution corresponds to $2N$ pixels at full resolution.
- 3 At each finer level, refine with a small ± 2 search centered on that doubled estimate, correcting only for the rounding introduced by downsampling.

Because the recursion bottoms out immediately for small images, this same function reproduces the single-scale results above on the low-res JPEGs — no separate code path was needed.

04 • Results

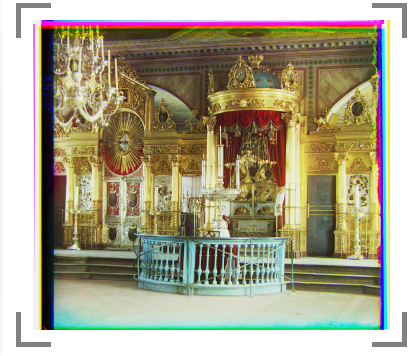
Pyramid results, full-resolution scans

Offsets computed by the pyramid search on every provided example image.



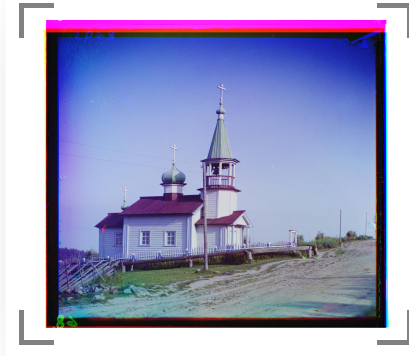
church

L2 dx +4, dy +25 (G) ·
dx -4, dy +58 (R) ·
NCC dx +4, dy +25 (G) ·
dx -4, dy +58 (R)



emir

L2 dx +24, dy +49 (G) ·
dx -168, dy +5 (R) ·
NCC dx +24, dy +49 (G) ·
dx -168, dy +5 (R)



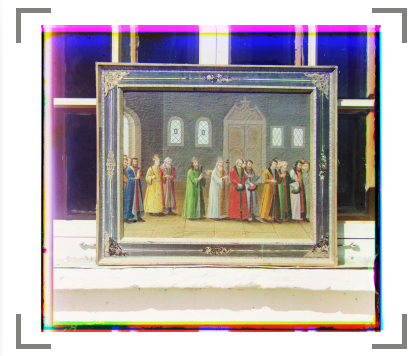
harvesters

L2 dx +17, dy +59 (G) ·
dx +13, dy +123 (R) ·
NCC dx +17, dy +59 (G) ·
dx +13, dy +123 (R)



icon

L2 dx +17, dy +41 (G) ·
dx +23, dy +89 (R) ·
NCC dx +17, dy +41 (G) ·
dx +23, dy +89 (R)



ilemselga

L2 dx +7, dy +40 (G) ·
dx +11, dy +131 (R) ·
NCC dx +7, dy +40 (G) ·
dx +11, dy +131 (R)



melons

L2 dx +10, dy +81 (G) ·
dx +12, dy +174 (R) ·
NCC dx +10, dy +81 (G) ·
dx +12, dy +174 (R)



religious_painting

L2 dx +2, dy +28 (G) ·
dx +7, dy +69 (R) ·
NCC dx +2, dy +28 (G) ·
dx +7, dy +69 (R)



self_portrait

L2 dx +29, dy +78 (G) ·
dx +36, dy +174 (R) ·
NCC dx +29, dy +78 (G) ·
dx +36, dy +174 (R)



siren

L2 dx -6, dy +49 (G) ·
dx -24, dy +96 (R) ·
NCC dx -6, dy +49 (G) ·
dx -24, dy +96 (R)

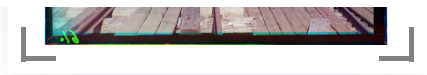




three_generations

L2 dx +13, dy +52 (G) ·
dx +11, dy +111 (R)

NCC dx +13, dy +52 (G) ·
dx +11, dy +111 (R)



wharf

L2 dx -7, dy +15 (G) ·
dx -16, dy +82 (R)

NCC dx -7, dy +15 (G) ·
dx -16, dy +82 (R)

05 · Results

Additional examples from the collection

Three more plates downloaded directly from the [Library of Congress collection](#), run through the same pyramid pipeline.



bashkir_woman_in_a_folk_costume

L2 dx -5, dy +60 (G) ·
dx -21, dy +130 (R)

NCC dx -5, dy +60 (G) ·
dx -21, dy +130 (R)



john_the_theologian

L2 dx +16, dy +20 (G) ·
dx +17, dy +59 (R)

NCC dx +16, dy +20 (G) ·
dx +17, dy +59 (R)



at_harvest_time

L2 dx +27, dy +24 (G) ·
dx +39, dy +144 (R)

NCC dx +27, dy +24 (G) ·
dx +39, dy +144 (R)

06 · Where it broke

The Emir of Bukhara

`emir.tif` is the one image the assignment itself flags as a hard case — the three exposures don't share the same brightness relationships the way they do in the other plates, so a metric built on raw pixel values (both L2 and NCC here) isn't guaranteed to find the shift that's actually correct.



L2 · dx +24, dy +49 (G)



NCC · dx +24, dy +49 (G)

The green channel converges cleanly and the robe, floor, and doorway all align well under both metrics. The red channel is the harder problem: a narrow search window pins the result to its own edge (evidence the true shift lies further out), but widening the window lets the search wander to a shift that scores better under raw pixel L2/NCC while visibly *worsening* alignment across the whole frame — a false optimum, not a real improvement. What's left in the images above is a single ghosted outline around the Emir's head and shoulder, most likely because he shifted slightly during the several-second gap between the sequential color exposures — something no global translation can correct, regardless of the metric used to find it.

This is the one image left imperfectly aligned in the final results.